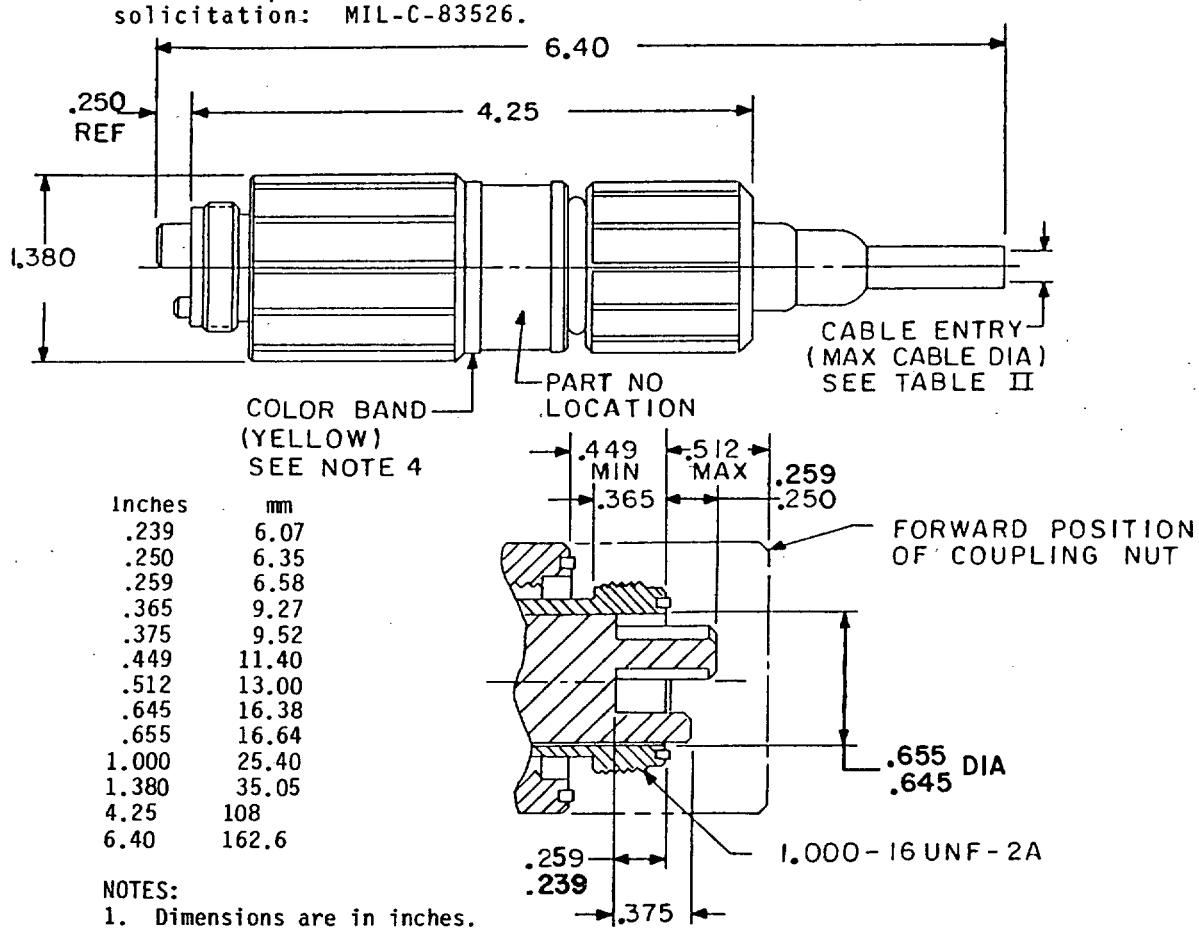
MILITARY SPECIFICATION SHEET

CONNECTORS, PLUG, FIBER OPTIC, CIRCULAR, HERMAPHRODITIC, IN-LINE MOUNTING, 2 AND 4 POSITIONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-83526.



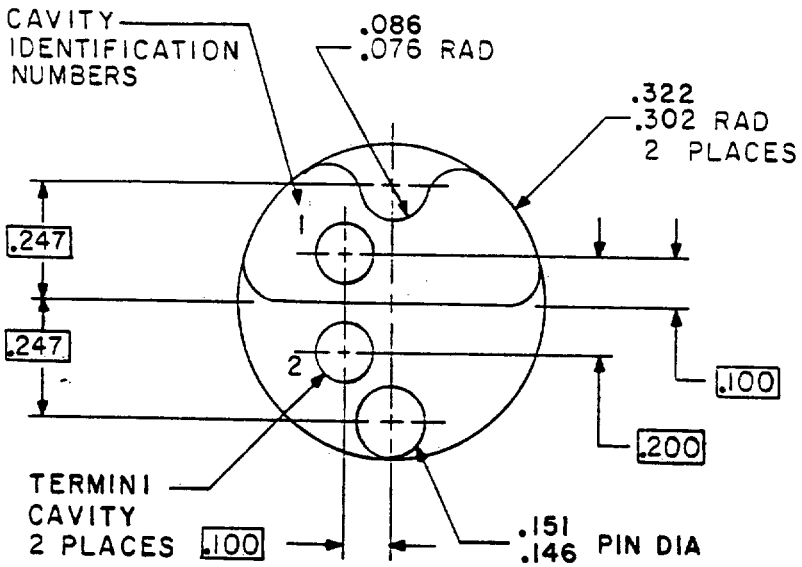
Inches	mm
.239	6.07
.250	6.35
.259	6.58
.365	9.27
.375	9.52
.449	11.40
.512	13.00
.645	16.38
.655	16.64
1.000	25.40
1.380	35.05
4.25	108
6.40	162.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals and ±.03 (0.8 mm) for two place decimals. Angular tolerances are ±0° 30'.
4. The phrase "Fiber-Optic" must be included within the yellow band.
5. Unless otherwise specified, dimensions are symmetrical about centerlines.

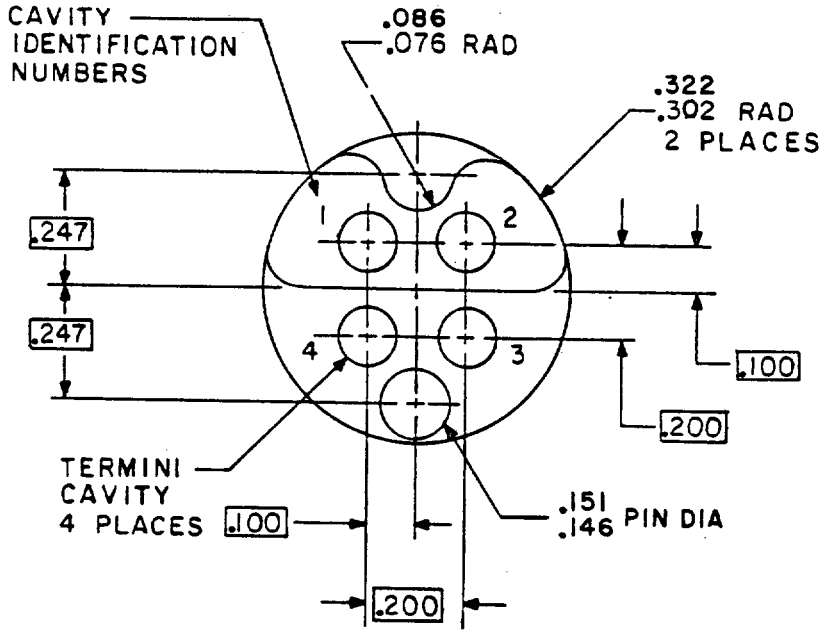
FIGURE 1. In-line connector.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2 Termini connector type 1-01



Inches	mm
.076	1.93
.086	2.18
.100	2.54
.146	3.71
.151	3.84
.200	5.08
.247	6.27
.302	7.67
.322	8.18

4 Termini connector type 1-02

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Terminal configuration face view.

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REQUIREMENTS:

Design and construction:

Dimensions and configurations: See figures 1 and 2.

Fiber optic cable:

Cable diameter: In accordance with DOD-C-85045.

Cable configuration: In accordance with DOD-C-85045.

Fiber diameter: In accordance with DOD-F-49291.

Fiber numerical aperture: In accordance with DOD-F-49291.

Fiber type: Multimode.

Fiber class: Graded index.

Fiber composition: Glass/glass.

Materials: The material requirements are listed in table I.

Connector termination and assembly tools: As specified in MIL-K-83533.

Cleaning procedures: The cleaning procedures for the connector shall be included with the assembly instructions of the connector.

Termini: The required number of termini shall be supplied with each connector unless otherwise specified (see table II). Style P(pin) or S(socket) MIL-T-29504/1 and MIL-T-29504/2. For dummy terminus, see MIL-T-29504/3.

Weight: 8.0 ounces maximum.

Coupling forces: 11.0 pounds.

Coupling torques: 5 inch-pounds minimum.

Identification marking:

Part or Identifying Number (PIN): See table III.

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TABLE I. Material description.

Parts description	Material	Finish
Spacer Connector shell Insert (2 termini) Insert (4 termini) Compression nut Coupling nut	Aluminum alloy 2011-T3 QQ-A-225/3 " "	Anodized MIL-A-8625 type II, class 2 Color: Gray
Strain relief Compression sleeve Strain relief sleeve	" " "	Chemical film MIL-C-5541 class 1A
Dowel pins Securing screw	Cres type 303 or 303SE per ASTM A581/A582 condition A	Passivated MIL-S-5002
Strain relief boot O-rings	Fluorosilicone rubber MIL-R-25988 type I, class 1	None None
Shrink tubing	Melt seal polyolefin MIL-I-23053/4	None
S/H cap screw	MS16995-9	None

PIN:

M83526/01- 01 X A, B, or N

Basic specification

Terminus insert arrangement

Cable entry diameter (see table II)

Connector supplied with (A) 50/125 size termini,
(B) 100/140 size termini, or (N) without termini

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TABLE II. Cable entry diameter and PIN designation.

Cable entry diameter (over jacket)	Designation number
.216	1
.255	2
.330	3
.390	4

PIN: M83526/01- (dash number from table III). Marked on connector.

TABLE III. Parts and characteristics.

PIN M83526/01-	Insert arrangement	Cable entry diameter	Cable PIN D85045	Fiber size DOD-F-49291	Termini supplied with connector
011A	2 termini insert	.216	B2A	50/125	Yes
021A	4 termini insert	.216	1/	50/125	Yes
011B	2 termini insert	.216	C2A	100/140	Yes
021B	4 termini insert	.216	1/	100/140	Yes
011N	2 termini insert	.216	N7A	N/A	No
021N	4 termini insert	.216	N/A	N/A	No
012A	2 termini insert	.255	1/	50/125	Yes
022A	4 termini insert	.255	B4A	50/125	Yes
012B	2 termini insert	.255	1/	100/140	Yes
022B	4 termini insert	.255	C4A	100/140	Yes
012N	2 termini insert	.255	N/A	N/A	No
022N	4 termini insert	.255	N/A	N/A	No
013A	2 termini insert	.330	1/	50/125	Yes
023A	4 termini insert	.330	1/	50/125	Yes
013B	2 termini insert	.330	1/	100/140	Yes
023B	4 termini insert	.330	1/	100/140	Yes
013N	2 termini insert	.330	N7A	N/A	No
023N	4 termini insert	.330	N/A	N/A	No
014A	2 termini insert	.390	1/	50/125	Yes
024A	4 termini insert	.390	1/	50/125	Yes
014B	2 termini insert	.390	1/	100/140	Yes
024B	4 termini insert	.390	1/	100/140	Yes
014N	2 termini insert	.390	N7A	N/A	No
024N	4 termini insert	.390	N/A	N/A	No

1/ D85045 cable not available for these connectors.

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Quality conformance and periodic inspection: The following cables shall be used for quality conformance and periodic inspection.

Number channels	Cables
2	PIN M85045/2-B2A, 50 micron core, 125 clad, or equal
4	PIN M85045/2-C4A, 100 micron core, 140 clad, or equal.

For quality conformance and periodic inspection, dummy termini shall be used in all unused cavities.

Patent notice: The Government does not have royalty-free license under the following patent for the benefit of manufacturers of the item, either for the Government or for use in equipment to be delivered to the Government.

Patent number	Patent expiration date
US 3,994,564	11/30/1993
US 4,054,366	10/18/1994
US 4,060,308	11/29/1994

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, SH
Air Force - 11, 17, 19, 80, 90, 99
DLA - ES

User activities:
Navy - MC, OS, YD
Air Force - 13, 14

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0065-1)

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